

B.Tech. Project End Sem Exam Presentation on

Cyclone Detection and Intensity Estimation using Satellite Images

Presented by

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Project Guide

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Why Cyclone Detection?

- Cyclones are highly destructive natural disasters, particularly affecting coastal areas of India.
- Over 117 cyclones impacted India between 1970 and 2019, causing 40,000+ deaths.
- Early detection can save lives and minimize economic loss.
- Effective tracking is vital for disaster preparedness and timely evacuation.

Why Cyclone Detection advancement is required?

- Traditional systems rely heavily on human interpretation, which is slow and subjective.
- Manual approaches often delay warnings and limit large-scale automation.
- There's a growing need for real-time, automated systems that reduce human error.

- Advanced technology enables higher accuracy and quicker response times.

Traditional Methods for Cyclone Detection

- Dvorak Technique: Relies on visual pattern recognition from IR satellite images.
- Kalman Filters and statistical methods for cyclone genesis prediction.
- Limitations:
 - Time-consuming and not scalable.
 - High false positive rate and poor generalization across regions.
 - Requires skilled meteorologists to interpret data manually.

Introduction Contd...

Early studies pioneered the use of satellite data for cyclone genesis detection and center localization

Subsequent advancements

- Early studies pioneered the use of satellite data for detecting cyclone genesis and center localization.
- Introduction of Convolutional Neural Networks (CNNs) enabled automated cyclone intensity estimation.
- Hybrid models were developed for multi-sensor satellite data fusion to improve accuracy.
- Advanced deep learning architectures like Faster R-CNN and YOLOv8 were applied for real-time cyclone tracking.
- Use of super-resolution techniques enhanced feature extraction from low-resolution satellite imagery.

Overview of Proposed Work

- Developed an automated cyclone detection system using CNN models.
- Dataset: INSAT-3D IR satellite images (2013–2023) sourced from Kaggle/MOSDAC.
- Web scraping used to fetch real-time satellite data.
- Image preprocessing includes enhancement and normalization.
- CNN predicts intensity based on extracted features.
- Integrated with a Flask API and React dashboard for real-time visualization.
- Final output: cyclone alerts, intensity predictions, and live weather map interface.

Traditional Methods & Early Satellite Data Utilization (Pre-2010s)

- N Uma et al., Dvorak Technique, Advanced Dvorak Technique (ADT), and Passive Microwave Imagery: Manual analysis of cloud patterns and thermal infrared (IR) data.
- Piñeros et al., Kalman Filters and statistical approaches like axisymmetric cloud organization analysis were used for cyclone tracking and genesis prediction, achieving 93% true positive rates but with high false alarms.
- Talukder Ashit et al., QuikSCAT (wind data) and TRMM (precipitation data) enabled global cyclone detection, achieving a 91.67% true positive rate.

Machine Learning Integration (2010s)

- MD. Haque Nazmul et al., Used support vector machines (SVM), decision trees (DT), and random forest (RF) achieved 86–95% accuracy in cyclone classification.
- The ISCCP and NOAA GridSat dataset standardized geo-stationary satellite data (1980–2011) for climate studies, enabling global cyclone tracking.
- Shakya et al., Linear regression and optical flow techniques improved cyclone path forecasting and temporal resolution.

Early Deep Learning Adoption (Late 2010s–Early 2020s)

- Kavitha D., CNNs emerged for spatial feature extraction, achieving 95% detection accuracy using IR/WV satellite imagery
- Suresh Anjana, Hybrid CNN-RNN models combined spatial and temporal data, reducing RMSE to 3.58 knots
- YANG Wen et al., Pre-trained CNNs (VGGNet, ResNet) achieved a 4.05 m/s MAE in intensity estimation, outperforming the Dvorak technique
- Xie et al., YOLOv4 and Faster R-CNN were applied for cyclone eye localization, achieving 87.86% mAP

Advance Deep Learning Adoption (2020s - 2024s)

- Shakya et al., RetinaNet and LSTM hybrid models achieved 97% classification accuracy with optical flow interpolation.
- Hybrid CNN + YOLOv3 models achieved 98.4% binary classification accuracy for North Indian Ocean cyclones. [An End-to-End Deep Learning Framework, 2024]
- Xie et al., Data fusion of ASCAT wind fields and precipitation radar data enhanced global detection accuracy using FPN + Faster R-CNN.
- Chang-Jiang Zhang et al., Global hybrid models using GOES-16 satellite imagery achieved 94% accuracy 12 hours before landfall

Research Gap Analysis

- 1 Many existing studies have advanced models like ViTs achieve high accuracy but require significant computational resources, limiting real-time deployment.
- 2 High-resolution satellite systems are costly and inaccessible for developing regions.
- 3 Places like the North Indian Ocean have fewer cyclone records, making AI models less accurate there.
- 4 Early cyclone detection systems often mistake normal storms for cyclones (false alarms).

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Traditional methods for cyclone detection often rely on manual analysis and heuristic approaches, which can be slow and inaccurate. Our study shows for more automated and efficient cyclone detection systems that take advantage of the vast amounts of satellite imagery and meteorological data available. using deep learning approach like CNN for better cyclone intensity estimation.

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- 1 To make a survey of existing techniques that are proven sound enough to cyclone detection
- 2 Develop a Deep Learning-Based Cyclone Detection System.
- 3 Improve the prediction of the intensity Estimation as well as wind speed detection..
- 4 Implement a real-time weather map to monitor and visualize wind speed patterns.

① Data Collection & Preprocessing

- Data Retrieval
 - web scraping to collect satellite imagery
 - Image Data - INSAT 3D satellite
- Preprocessing:
 - Normalize, enhance, and clean the satellite imagery.

② Model Development

- convolutional neural network (CNN)

3 Training & Validation:

- CNN on historical cyclone data on 80% data.
- Testing: on 20% of original data.

4 Backend Development (Flask):

- Takes cyclone satellite images,
- Predicts cyclone intensity using a pretrained convolutional neural network (CNN),
- If *intensity* > 34 $\rightarrow$ "99% chances that cyclone will hit.

5 Frontend Implementation (Flask Template)

- HTML + Jinja2 (Flask's template engine).
- CSS for styling.

Proposed Methodology Contd...

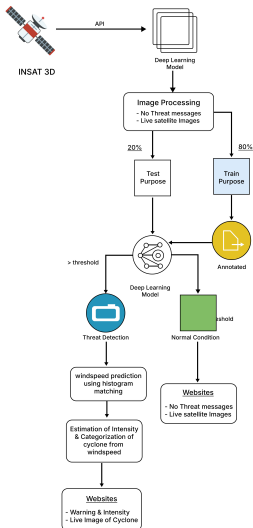
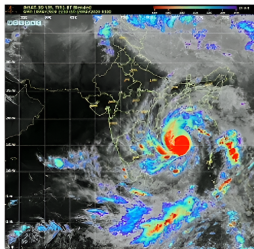


Figure 1: Proposed Workflow for Cyclone Detection

The live images captured by INSAT-3D Satellite were retrieved.

Cyclone



No Cyclone

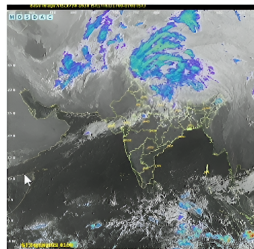
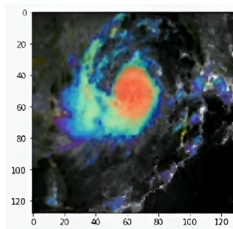


Figure 2: IR Images captured by INSAT-3D

Proposed Methodology Contd...

The important features of the image were extracted using dilation and erosion before passing it into the model. This step is necessary so as to improve the model performance.

Cyclone



No Cyclone

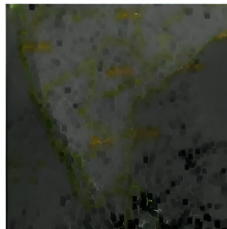


Figure 3: Image Extraction

Proposed Methodology Contd...

After extracting the important features the images were passed through CNN model which distinguished the low pressure region by turning it into violet colour and the background into yellow.

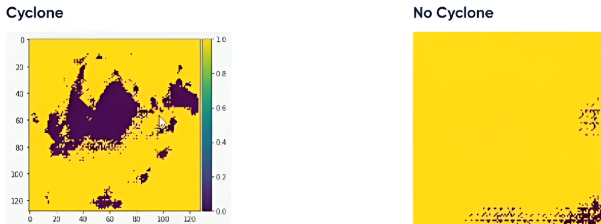


Figure 4: Pressure Extraction from image

● Frontend:

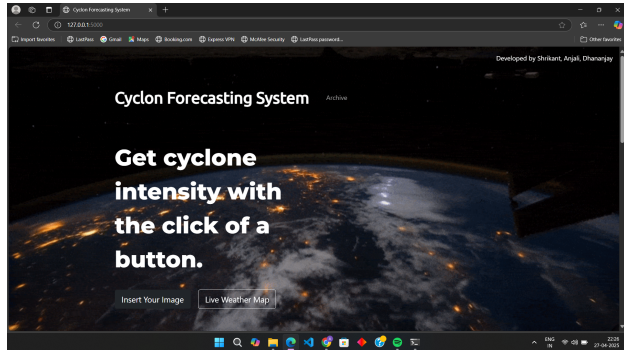


Figure 5: Main Dashboard - Cyclone Forecasting System

● Frontend Interface Components:

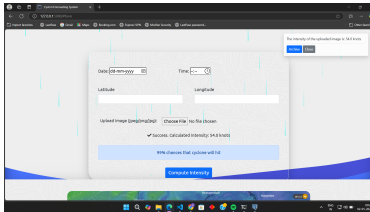


Figure 6: Cyclone Intensity Estimation

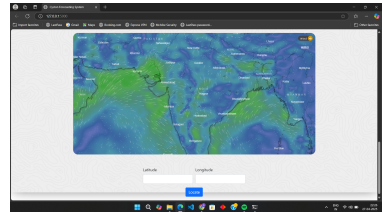


Figure 7: Real-Time Wind Path Monitoring

- Figure 6 shows the result of the cyclone intensity estimation.
- Figure 7 shows the Real-Time Wind Path Monitoring.

● Backend Components:

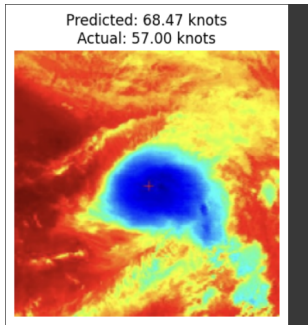


Figure 8: Predicted VS Actual Intensity

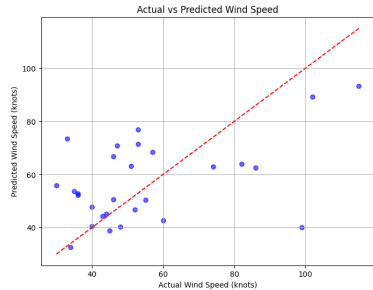


Figure 9: Graph Predicted vs Actual

- Model prediction: 68.47 knots
- Actual Intensity : 57.00 knots

Result and Discussion Contd...

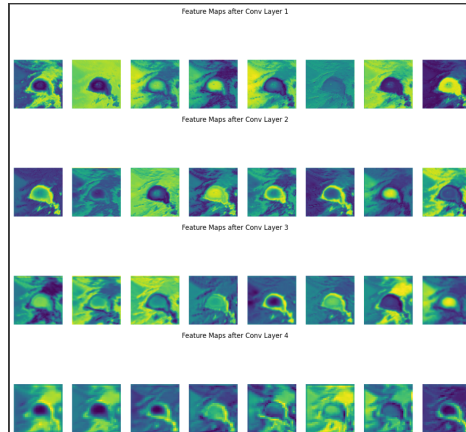


Figure 10: CNN Layers

- Feature maps after CNN layer used for model training.

Result and Discussion Contd...

- If the Wind speed was found with correlation of histogram data between test and trained image is 119 knots.

Serial No.	T number	Classification of Cyclonic Disturbance	Wind Speed	Wind Criteria	Pressure depth
1	T1.0	Low	-	<17	-
2	T1.5	Depression	25	17-27	-
3	T2.0	Deep Depression	30	28-33	4.5
4	T2.5	Cyclonic Storm	35	34-47	6.1
5	T6.0	Extremely Severe Cyclonic Storm	115	92-119	65.6
6	T6.5	Super Cyclonic Storm	127	> 120	80

Figure 11: Cyclonic Disturbance Classification

Result and Discussion Contd...

Table 1: Model Performance Metrics for Cyclone Intensity Prediction

Metric	Value
Test RMSE	19.85 knots
Test MAE	15.44 knots
Test MSE	393.90 knots
R-squared (R^2)	0.19

Interpretation:

- **RMSE** gives higher weight to large errors
- **MAE** represents average error magnitude
- **R^2** indicates only 19% variance explained

Conclusion

- CNN-based detection shows promising results for tropical cyclone intensity estimation.
- Enables real-time response, especially useful for coastal disaster preparedness.
- System integrates prediction, monitoring, and UI in one platform.
- Future enhancements needed for:
 - Reducing false alarms,
 - Improving generalization across geographies,
 - Faster runtime models.

Future Scope

- Improve accuracy using ensemble models and ViTs.
- Integrate multi-source data (surface pressure, wind vectors).
- Real-time alert system for government/emergency bodies.
- High-resolution forecasting models for localized prediction.
- Use AI-powered edge computing for remote sensor data processing.

Publication



Figure 12: Book Chapter: Coastal Environments - Risk Assessment, Restoration and Sustainable Solution

The book chapter titled "Tropical Cyclone Detection and Intensity Estimation" is communicated to the book "Coastal Environments: Risk Assessment, Restoration and Sustainable Solutions" published by Springer Nature under the Sustainability Solutions book series.

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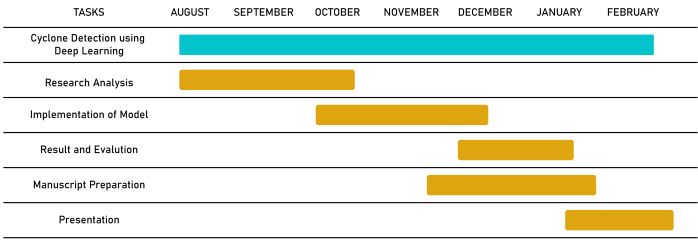


Figure 13: Timeline Chart

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Questions